HW 7

1) $Pr(\alpha_1, \ldots \alpha_n | \beta) = Pr(\alpha_1 | \alpha_2, \ldots, \alpha_n, \beta) Pr(\alpha_2 | \alpha_3 \ldots, \alpha_n, \beta) \ldots Pr(\alpha_n | \beta)$

def of cond probability ‖ apply chain rule

$$\frac{Pr(\alpha_1, \ldots \alpha_n, \beta)}{Pr(\beta)} = \frac{Pr(\alpha_1 | \alpha_2, \ldots \alpha_n, \beta) \cdots Pr(\alpha_n | \beta) \cdot Pr(\beta)}{Pr(\beta)}$$

$$= Pr(\alpha_1 | \alpha_2, \ldots \alpha_n, \beta) Pr(\alpha_2 | \alpha_3, \ldots, \alpha_n, \beta) \ldots Pr(\alpha_n | \beta)$$

this is the right side □

2) $Pr(O) = 0.5$ $Pr(P|O) = 0.9$ $= P(O|P) = ?$ $\frac{Pr(P|O) \cdot Pr(O)}{Pr(P)}$

$Pr(N) = 0.2$ $Pr(P|N) = 0.3$

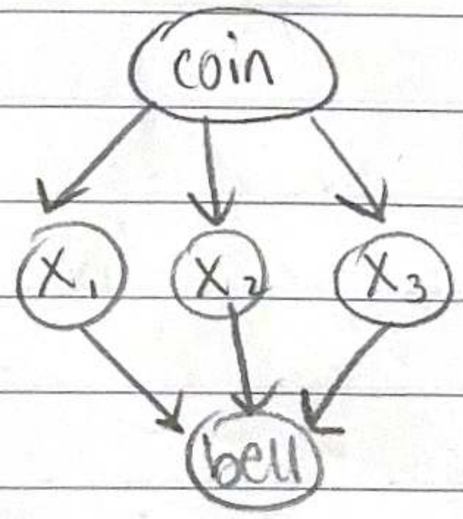
$Pr(T) = 0.3$ $Pr(P|T) = 0.1$

$$Pr(P) = Pr(P|O)Pr(O) + Pr(P|N)Pr(N) + Pr(P|T)Pr(T)$$
$$= (0.9 \times 0.5) + (0.3 \times 0.2) + (0.1 \times 0.3)$$
$$= 0.45 + 0.06 + 0.03$$
$$= 0.54$$

$$Pr(O|P) = \frac{(0.9)(0.5)}{(0.54)} = 0.833$$

83.3% that oil is present, given the circumstances in the problem

3) a 20%
b 40%
c 80%



coin CPT:

coin	Pr(coin)
a	1/3
b	1/3
c	1/3

X CPTs:

coin	X_1	$Pr(X_1 \mid coin)$
a	H	0.2
a	$\bar{H}$	0.8
b	H	0.4
b	$\bar{H}$	0.6
c	H	0.8
c	$\bar{H}$	0.2

coin	X_2	$Pr(X_2 \mid coin)$
a	H	0.2
a	$\bar{H}$	0.8
b	H	0.4
b	$\bar{H}$	0.6
c	H	0.8
c	$\bar{H}$	0.2

coin	X_3	$Pr(X_3 \mid coin)$
a	H	0.2
a	$\bar{H}$	0.8
b	H	0.4
b	$\bar{H}$	0.6
c	H	0.8
c	$\bar{H}$	0.2

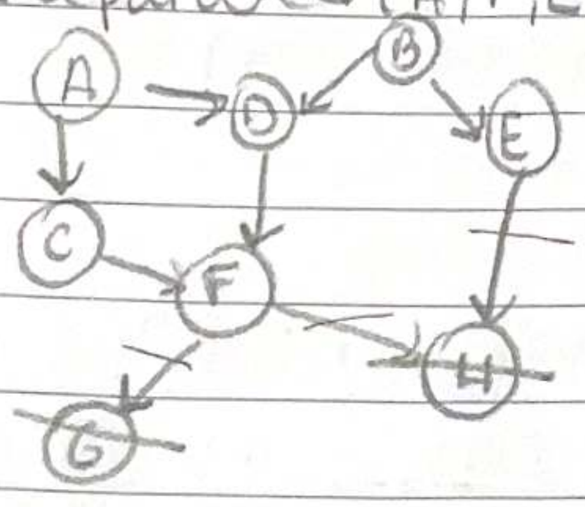
Bell CPT:

Bell	X_1	X_2	X_3	$Pr(Bell \mid X_1, X_2, X_3)$
Y	H	H	H	1
N	H	H	H	0
Y	$\bar{H}$	H	H	0
N	$\bar{H}$	H	H	1
Y	H	$\bar{H}$	H	0
N	H	$\bar{H}$	H	1
Y	H	H	$\bar{H}$	0
N	H	H	$\bar{H}$	1
Y	$\bar{H}$	$\bar{H}$	H	0
N	$\bar{H}$	$\bar{H}$	H	1
Y	H	$\bar{H}$	$\bar{H}$	0
N	H	$\bar{H}$	$\bar{H}$	1
Y	$\bar{H}$	H	$\bar{H}$	0
N	$\bar{H}$	H	$\bar{H}$	1
Y	$\bar{H}$	$\bar{H}$	H	1
N	$\bar{H}$	$\bar{H}$	H	0

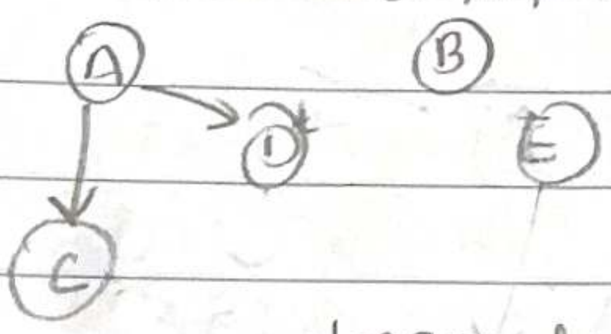
4.

	Parents	Descendants(v)	Non-descendants(v)	a)
A:	–	C, D, F, G, H	B, E	I(A, ∅, BE)
B:	–	D, E, F, G, H	A, C	I(B, ∅, AC)
C:	A	F, G, H	B, D, E	I(C, A, BDE)
D:	A, B	F, G, H	C, E	I(D, AB, CE)
E:	B	H	A, C, D, F, G	I(,E, B, ACDFG)
F:	C, D	G, H	A, B, E	I(F, CD, ABE)
G:	F	–	A, B, C, D, E, H	I(G, F, ABCDEH)
H:	E, F	–	A, B, C, D, G	I(H, EF, ABCDG)

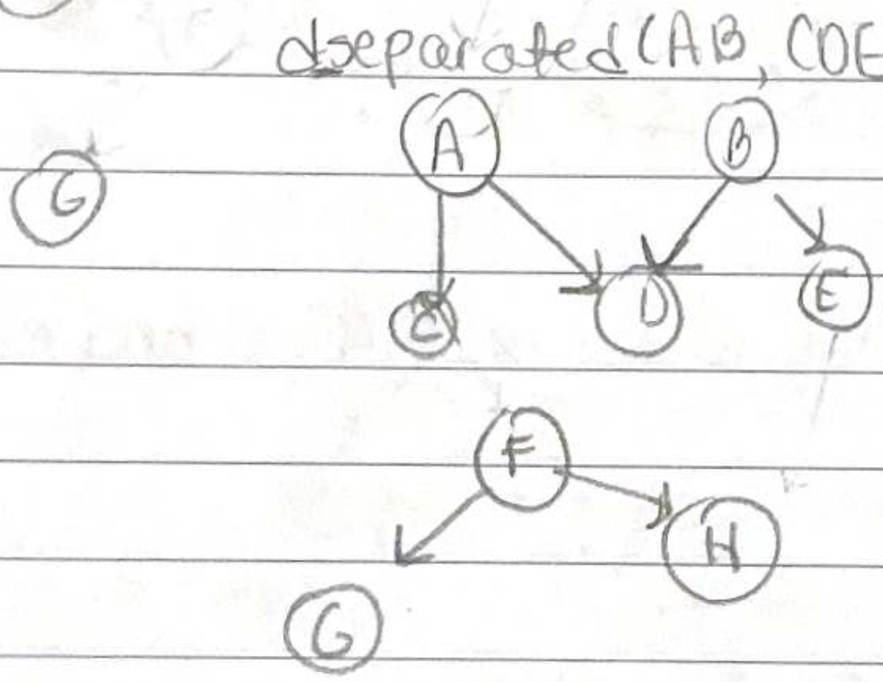
b) d_separated(A, F, E) Since A + E are not disconnected, d-sep(A, F, E) is false.



d-separated(G, B, E) G + E are disconnected, so this is true.



dseparated(AB, COE, GH) True, since AB + GH are disconnected.



c) $Pr(a,b,c,d,e,f,g,h) = Pr(a) \cdot Pr(b) \cdot Pr(c|a) \cdot Pr(d|a,b) \cdot Pr(e|b) \cdot Pr(f|cd) \cdot Pr(g|f) \cdot Pr(h|ef)$

d) $Pr(A=1, B=1)$ A and B are independent, both root nodes $I(A, \emptyset, BE)$

$= Pr(A=1) \cdot Pr(B=1) = .2 \times .7 = \boxed{0.14}$

$Pr(E=0|A=0)$ $I(E, B, ACDFG)$ E + A ind, given B (ind)

$= Pr(E=0|B=0) \cdot Pr(B=0|A=0) + Pr(E=0|B=1) \cdot Pr(B=1|A=0)$ (ind)

$= Pr(E=0|B=0) \cdot Pr(B=0) + Pr(E=0|B=1) \cdot Pr(B=1)$

$= (.1) \times (.3) + (.9) \times (.7)$

$= .03 + .63 = \boxed{.66}$

5) a) $\{w_0, w_2, w_3\}$ b) $Pr(\alpha) = Pr(w_0) + Pr(w_2) + Pr(w_3) = 0.3 + .1 + 0.4 = \boxed{0.8}$

c) $P(A,B|\alpha) = \frac{Pr(A,B \wedge \alpha)}{Pr(\alpha)} = \frac{Pr(w_0)}{0.8} = \frac{0.3}{0.8} = \boxed{0.375}$

d) $Pr(A \Rightarrow \neg B|\alpha) = Pr(\neg A \vee \neg B|\alpha) = \frac{Pr((\neg A \vee \neg B) \wedge \alpha)}{Pr(\alpha)} =$

$\frac{0.1 + 0.4}{0.8} = \frac{0.5}{0.8} = \boxed{0.625}$

w_2, w_3 are only worlds where α is True and either A or B is False.